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$$\begin{aligned}& \int \cos x \cosh x dx \\& \begin{cases} u = \cos x \\ dv = \cosh x dx \end{cases} \Rightarrow \begin{cases} du = -\sin x dx \\ v = \sinh x \end{cases} \\& \Rightarrow \cos x \sinh x - \int \sin x \sinh x dx \\& \begin{cases} u = \sin x \\ dv = \sinh x dx \end{cases} \Rightarrow \begin{cases} du = \cos x dx \\ v = \cosh x \end{cases} \\& \Rightarrow \cos x \sinh x + \sin x \cosh x - \int \cos x \cosh x dx \\& 2I = \cos x \sinh x + \sin x \cosh x + C \\& \Rightarrow I = \frac{1}{2}(\cos x \sinh x + \sin x \cosh x) + C\end{aligned}$$

References